

Midterm 1

● Graded

Student

James Stuart

Total Points

98 / 100 pts

Question 1

Question 1

30 / 30 pts

1.1 a

15 / 15 pts

✓ - 0 pts Correct

- 3 pts first one is wrong

- 3 pts second is wrong

- 3 pts third is wrong

- 3 pts fourth is wrong

- 3 pts fifth is wrong

1.2 b

7 / 7 pts

✓ - 0 pts Correct

- 1 pt unit is wrong (should be 6000 kJ)

- 1 pt minutes to seconds conversion

- 1 pt watts= joules/seconds

- 7 pts not done or wrong

1.3 c

8 / 8 pts

✓ - 0 pts Correct

- 0 pts Pressure = ρgh

- 8 pts not done or wrong

- 1 pt unit is wrong

- 1 pt ρ is wrong

- 1 pt needs to determine the difference

Question 2

Question 2

33 / 35 pts

2.1 — a

2 / 2 pts

✓ - 0 pts Correct

2.2 — b

4 / 4 pts

✓ - 0 pts Correct

- 1 pt 3-1 is wrong

- 4 pts wrong or not done

- 1 pt not water, this is air. Not dome drawing

- 1 pt x axis should be specific volume, not volume.

2.3 — c,d

5 / 5 pts

✓ - 0 pts Correct

- 1 pt closed system

- 1 pt $\Delta U = Q - W$

2.4 — e

14 / 15 pts

- 0 pts Correct

- 5 pts W1-2 is wrong

- 4 pts W2-3 is wrong

- 6 pts W3-1 is wrong

- 1 pt W1-2 should be negative

- 2 pts n value for polytropic process is wrong

- 2 pts unit conversion

- 1 pt W3-1 should be positive

- 1 pt W3-1 unit conversion

- 3 pts W3-1 should be calculated by natural log

✓ - 1 pt W 3-1 calculation error

- 1 pt calculation error W1-2

2.5 

8 / 9 pts

– 0 pts Correct

✓ – 1 pt calculation error

– 9 pts not done or wrong

Question 3

Question 3

35 / 35 pts

3.1 a

2 / 2 pts

✓ - 0 pts Correct

3.2 b

6 / 6 pts

✓ - 0 pts Correct

- 3 pts domes are missing on the graphs

- 5 pts states are not labeled on the graph

- 1 pt state 1 is wrong on Pv graph

- 1 pt state 1 is wrong on Tv graph

- 1 pt state 2 is wrong on Pv graph

- 1 pt state 2 is wrong on Tv graph

- 1 pt state 3 is wrong on Pv graph

- 1 pt state 3 is wrong on Tv graph

3.3 c,d

5 / 5 pts

✓ - 0 pts Correct

- 5 pts not done or wrong

- 1 pt closed system

3.4 e

10 / 10 pts

✓ - 0 pts Correct

- 10 pts not done or wrong

- 6 pts W2-3 is not done or wrong (should be constant P)

- 1 pt calculation error

- 1 pt W2-3 should be negative

- 1 pt P is wrong

- 2 pts v3 is wrong

- 2 pts v2 is wrong

3.5 f

10 / 10 pts

✓ - 0 pts Correct

- 1 pt calc erro

- 3 pts x not calculated or wrong

- 3 pts u1 is wrong

- 10 pts not done

- 1 pt u3 is wrong

- 2 pts deltaU calculations are wrong

3.6 g

2 / 2 pts

✓ - 0 pts Correct

- 1 pt state 1 is wrong

- 1 pt state 2 is wrong

- 2 pts not done or wrong

- 0.5 pts state 3 is wrong

CHE 201 MidTerm Exam 1

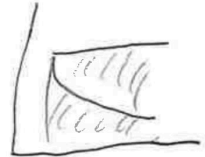
September 29th, 2025

NAME: James Stuart

Please fill in answers in the space supplied and show all your work to receive full credit.

1. a) (15 pts) State whether the following statements are True or False

F An ideal gas at a given state expanded to a fixed final volume at constant pressure or at constant temperature. Value of work is smaller in constant pressure case.



T Seven people stand on a scale and have a mass of 1280 lb. A specific property called the pep is defined as the number of people divided by mass. Pep is an intensive property.

F Isochoric process means no heat transfer.

F A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality more than 0.5. *(qual > 0.5 means vapor < liquid)*

F Energy transferred by heat to a closed system can be calculated by the area under the curve in P-V diagram.

b) (7pts) (FE Exam)

A 2-kW electric resistance heater in a room is turned on and kept on for 50 min. Determine the amount of energy transferred.

$$\frac{2 \text{ kW}}{1} \times \frac{60 \text{ s}}{1 \text{ min}} \times 50 \text{ min} = \boxed{6000 \text{ kJ}}$$

c) (8pts) (FE Exam)

Consider a fish swimming 5 m below the free surface of water. Determine the increase in the pressure exerted on the fish when it dives to a depth of 25 m below the free surface.

$$P_{\text{hydro 1}} = \frac{1000 \text{ kg}}{\text{m}^3} \cdot \frac{9.8 \text{ N}}{\text{kg}} \cdot 5 \text{ m} = 49000 \frac{\text{N}}{\text{m}^2} = 49 \text{ kPa}$$

$$P_{\text{hydro 2}} = \frac{1000 \text{ kg}}{\text{m}^3} \cdot \frac{9.8 \text{ N}}{\text{kg}} \cdot 25 \text{ m} = 245000 \frac{\text{N}}{\text{m}^2} = 245 \text{ kPa}$$

$$\rho = \frac{1 \text{ g}}{\text{cm}^3} \cdot \left(\frac{1000 \text{ cm}^3}{1 \text{ m}^3}\right)^3 \times \frac{1 \text{ (N)}}{1000 \text{ g}} = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$P = \rho g h$$

$$\Delta P = P_2 - P_1 = \boxed{196 \text{ kPa increase}}$$



2. (35 points)

A gas in a piston-cylinder assembly (7kg) at 5 bar and 15 m³ undergoes a cycle with three processes in series.

Process 1-2: Isobaric compression to 10 m³.

Process 2-3: Isochoric increase of pressure to 7.5 bar.

Process 3-1: Polytropic expansion to initial state.

Kinetic and potential effects can be ignored.

a) (2 pts) Draw a schematic of the system and label the system boundary.

b) (4 pts) Carefully sketch the process on a P-v diagram (Pressure-specific volume)

c) (3 pts) State the engineering model

d) (2 pts) Write the simplified energy balance

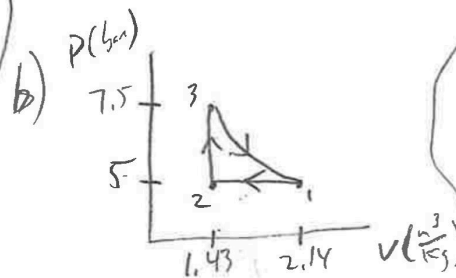
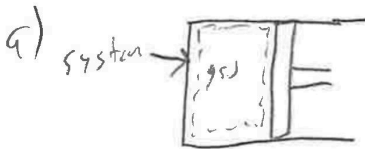
e) (15 pts) Calculate the work transferred in kJ for each step. Is work done on the system or by the system?

f) (9 pts) Calculate the overall heat transferred in kJ. Is heat transferred to the system or from the system?

$$v_1 = \frac{V_1}{m} = \frac{15 \text{ m}^3}{7 \text{ kg}} = 2.14 \text{ m}^3/\text{kg}$$

$$v_2 = \frac{V_2}{m} = 1.43 \text{ m}^3/\text{kg}$$

$$v_3 = 1.43 \text{ m}^3/\text{kg}$$



c) closed system
 $\Delta PE = \Delta KE = 0$
 process 1-2 is isobaric
 2-3 is isochoric
 3-1 is polytropic

d) $\Delta E = Q - W$

e) $W_{1-2} = (5 \text{ bar})(10 \text{ m}^3 - 15 \text{ m}^3) = -25 \text{ bar m}^3$
 $W_{1-2} = -2500 \text{ kJ}$
 work done on system

$W_{2-3} = 0$, no ΔV

$PV^n = C \rightarrow P_1 V_1^n = P_3 V_3^n \rightarrow 5 \text{ bar} (15 \text{ m}^3)^n = 7.5 \text{ bar} (10 \text{ m}^3)^n$
 $\left(\frac{3}{2}\right)^n = 1.5 \rightarrow n = 1$
 $PV = C \rightarrow C = 75$
 $P = \frac{75}{V}$

$W = 75 \int_{15}^{10} \frac{1}{v} dv = 75 \left[\ln v \right]_{15}^{10} = 75 \ln(2) = 51.99 \text{ bar m}^3$
 $W_{3-1} = 5200 \text{ kJ}$
 work done by system

$W_{net} = 2700 \text{ kJ}$
 work done by system

f) $\Delta E = Q - W = 0$
 $Q = W$

$Q = 2700 \text{ kJ}$

Heat transferred to the system

3. (35 points)

5 kg of liquid-vapor mixture of water contained in a piston-cylinder assembly undergoes a process from an initial state where the temperature is 90°C.

- **Process 1-2** System is heated at constant volume to a saturated vapor state at 170°C.
- **Process 2-3** System is cooled at constant pressure to 120°C.

Kinetic and potential effects can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (6 pts) Carefully sketch the process on a:
 - P-v diagram (Pressure-specific volume)
 - T-v diagram (Temperature-specific volume)
- (3 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (10 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- (10 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- (2 pts) Calculate the mass of vapor in each state in kg.

State 1: liquid-vapor mix

State 2: saturated vapor

State 3: compressed liquid

c) closed system

$$\Delta PE = \Delta KE = 0$$

process 1-2 is isochoric
2-3 is isobaric

$$\Delta E = Q - W = \Delta U$$

d)

e)

$$v_2 = 0.2428 \text{ m}^3/\text{kg} \quad u_2 = 2576.5 \text{ kJ/kg}$$

$$v_1 = 0.248 \text{ m}^3/\text{kg}, \text{ at } 90^\circ\text{C} \quad x = 0.102$$

$$0.248 = x(2.361) + (1-x)(1.0360 \times 10^{-3})$$

$$u_1 = (0.102)(2494.5) + (0.898)(376.85) = 592.85 \text{ kJ/kg}$$

State 3: compressed liquid - saturated liquid approx

$$v_3 = 1.0603 \times 10^{-3} \text{ m}^3/\text{kg} \quad u_3 = 507.50 \text{ kJ/kg}$$

$$P_2 = P_{\text{sat}} \text{ at } 170^\circ\text{C} = 7.917 \text{ bar}$$

$$W_{1-2} = 0, \text{ no } \Delta U$$

$$W_{2-3} = (7.917 \text{ bar})(5 \text{ kg})(1.0603 \times 10^{-3} \text{ m}^3/\text{kg} - 0.248 \text{ m}^3/\text{kg}) = -9.77 \text{ bar}\cdot\text{m}^3$$

$$\Delta U = m(u_3 - u_1) = (5 \text{ kg})(507.50 \text{ kJ/kg} - 592.85 \text{ kJ/kg}) = -446.75 \text{ kJ}$$

$$f) -446.75 \text{ kJ} = Q - (-9.77 \text{ kJ}) \rightarrow Q = -1423.75 \text{ kJ}$$

heat transferred from system

work done on system

$$W = -977 \text{ kJ}$$

g) state 1: $m_{\text{vapor}} = (0.102)(5 \text{ kg}) = 0.51 \text{ kg}$

state 2: $m_{\text{vapor}} = 5 \text{ kg}$

state 3: $m_{\text{vapor}} = 0 \text{ kg}$

